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Energy Consumption Analysis and Load Prediction in Battery Powered Vehicle Traffic: A SUMO-Based Simulation on Custom Road Network

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Registration Year: 2020-21

Academic Year: 2020-21

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This thesis report was submitted in partial fulfillment of the Requirements
for the Bachelor of Science Degree

**In Computer Science and Engineering
Faculty of Engineering and Technology**

November 2025

Jashore University of Science and Technology

Jashore-7408, Bangladesh



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Dedication

This work is lovingly dedicated to my family, whose endless support, prayers, and encouragement have shaped every step of my academic journey. Their sacrifices and belief in my abilities have been the strongest motivation throughout this thesis.

I also dedicate this work to my respected supervisor sir, whose wisdom and guidance have helped me grow academically and personally. Lastly, I express heartfelt gratitude to my friends, who have always stood beside me—sharing knowledge, offering inspiration, and making this entire journey meaningful.

Acknowledgments

I would like to begin by expressing my sincere gratitude to my supervisor, **Monishanker Halder Sir**, Assistant Professor, Department of Computer Science and Engineering. His continuous guidance, helpful suggestions, and constant support have been invaluable throughout this thesis. His encouragement and mentorship truly shaped the quality and direction of my work.

My heartfelt thanks go to my friends and classmates for their cooperation, encouragement, and meaningful discussions, which supported me at many stages of this journey.

Lastly, I am deeply grateful to my family. Their love, patience, and belief in me kept me motivated from beginning to end. Without their support, completing this thesis would not have been possible.

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Abstract

The rapid increase of Battery Powered Vehicles (BPVs)—including easy bikes, e-rickshaws, and electric vans—has introduced a new form of electrical load on Bangladesh’s urban distribution grid. Most of these vehicles charge directly from low-capacity neighborhood transformers without any formal scheduling or dedicated charging infrastructure. As a result, local feeders in cities such as Khulna and Jashore frequently face unpredictable load spikes and voltage stress. This study aims to understand how BPV movement patterns translate into real charging demand and how future load can be forecasted for improved grid planning. To achieve this, a complete SUMO-based simulation framework was developed, including a custom road network, multiple EV routes, node-level charging stations, and detailed battery behavior modeling. Using traffic microsimulation and EV energy extensions, the system generates high-resolution data on vehicle speed, SOC depletion, charging events, route-wise energy use, and node-level load accumulation. These outputs were then processed and capturing how different routes, congestion levels, and mobility patterns affect energy consumption. Finally, machine learning-based forecasting techniques were applied to predict future charging loads at different nodes based on SOC trends, route flows, and historical demand profiles. The results show that charging load does not simply follow traffic volume; instead, it concentrates at a few critical nodes where BPV routes converge or where travel energy cost is high. Several nodes emerged as potential transformer-stress points due to repeated, clustered charging behavior. The forecasting model further demonstrated that these load peaks can be predicted reliably, offering timely insight for distribution planners. Overall, this research provides one of the first simulation-driven analyses of BPV energy consumption and charging load forecasting in Bangladesh. The proposed framework can help power utilities, city planners design safer, more efficient charging networks, reduce local overload risks, and support Bangladesh’s transition toward sustainable electric mobility.

Chapter 1

Introduction

Chapter 1: Introduction

In this study, we address the growing challenge of power consumption and charging-load distribution caused by battery-powered vehicles (BPVs)—including easy bikes, e-rickshaws, and electric vans—that dominate urban transport in Bangladeshi cities such as Jashore and Khulna. These vehicles heavily strain local distribution grids, as large numbers of BPVs charge directly from low-capacity neighborhood transformers without coordination. To bridge the gap created by the lack of formal charging infrastructure, structured data, and predictive tools, we develop a simulation-based analytical framework using SUMO to model BPV movement, energy use, and node-wise charging demand across a custom urban network. Unlike studies focused on controlled EV charging stations, our approach captures real-world BPV behavior such as stop-and-go traffic, frequent recharging, and clustered charging patterns. By generating a high-resolution dataset of speed, SOC, charging events, and energy consumption, and forecasting charging loads at different grid nodes, the framework identifies potential transformer stress points, peak-load zones, and areas needing dedicated charging infrastructure. This provides a scalable decision-support tool for power utilities and city planners to manage BPV integration, prevent localized grid overload, and design smarter, more resilient urban charging networks as Bangladesh advances toward widespread electric mobility.

1.1 Background

The rapid rise of electric vehicles is reshaping transportation and power distribution systems. Unlike traditional combustion vehicles, BPVs or EVs shift energy demand from fuel stations to the electrical grid, creating spatially and temporally varying loads. Understanding EV movement, energy usage, and charging behavior is essential for load forecasting, grid reliability, and optimizing charging infrastructure. A key difficulty observed in Bangladeshi cities is the **uncoordinated and unmonitored charging behavior** of EVs.

Vehicles frequently cluster around specific nodes or roadside charging points, drawing large amounts of energy during peak hours. This often causes local grid stress, voltage drops, and transformer overloading—issues that are already prevalent in rapidly growing cities like Khulna. Moreover, energy usage varies widely depending on the vehicle’s speed, congestion level, route selection, and driving behavior. Without proper modeling and forecasting of these parameters, urban authorities and power distributors face increasing difficulty in ensuring grid reliability, planning future charging stations, and controlling unbalanced load distribution. But this research work with Traffic simulation tools like **SUMO** help represent real urban mobility, while EV extensions in SUMO allow modeling battery behavior, charging events, and energy consumption. Integrating these outputs with load prediction models forms a powerful approach for analyzing energy impacts of EV adoption.

1.1.1 Definition

1.1.1.1 Battery-Powered Vehicle (BPV) or EV

An Electric Vehicle (EV) or BPV is a transportation mode powered entirely by rechargeable batteries and electric motors rather than internal combustion engines. EVs store electrical energy in lithium-ion or lead-acid batteries and convert this stored energy into mechanical power through electric drivetrains. In the context of Bangladesh—particularly in urban regions such as Khulna and Jashore—EVs include widely used low-cost electric three-wheelers (e.g., easy bikes and e-rickshaws). These vehicles are favored due to their low operating cost, absence of tailpipe emissions, and suitability for short-distance intra-city travel. Their electricity-based operation directly links transportation demand with local electrical load, making EV behavior a crucial factor in grid planning.

1.1.1.2 Load Prediction

Load prediction refers to the analytical process of estimating the future electrical demand on the power grid over a specific time horizon. For EV-centric studies, this includes forecasting the temporal and spatial distribution of charging activities resulting from vehicle movements, battery depletion, and charging station utilization patterns. Accurate load prediction helps

power utilities anticipate peak demand periods, prevent transformer overloading, schedule energy dispatch efficiently, and design charging infrastructures that can support growing EV adoption. In Bangladesh’s urban centers, where power distribution systems already experience stress during peak hours, EV load forecasting is essential for preventing local grid failures.

1.1.1.3 Traffic Microsimulation

Traffic microsimulation is a fine-grained, vehicle-level modeling technique that captures the detailed movement of individual vehicles over time. It represents speed, acceleration, deceleration, lane-changing behavior, route selection, stop-and-go cycles, and congestion interactions with high temporal precision. In EV studies, microsimulation becomes even more critical because vehicle motion directly influences energy consumption and battery depletion dynamics. By integrating microsimulation with EV models, researchers can generate realistic estimates of power consumption across different roads, times, and traffic conditions—ultimately supporting more accurate charging load analysis and forecasting.

1.1.1.4 State of Charge (SOC)

State of Charge (SOC) is a quantitative measure indicating the remaining energy stored in an EV’s battery, usually expressed as a percentage (0–100%). SOC decreases during driving due to propulsion energy usage and increases when the vehicle is plugged into a charging station. SOC is a fundamental variable for understanding EV behavior because it determines when and where a vehicle will require charging. In simulation environments such as SUMO, SOC logs provide time-step-level insights into real-time power consumption, enabling the development of detailed charging demand models and node-level load prediction profiles.

1.1.1.5 Charging Station

A charging station is an electricity supply point where EVs replenish their battery energy. In simulation-based studies, charging stations are modeled as infrastructure elements attached to specific network nodes or road segments. Each station is characterized by properties such as charging power, availability, placement strategy, and duration of charging sessions. In Bangladesh’s urban landscape, the siting of EV charging stations is often unplanned, leading

to local load spikes and uneven power distribution. Within the simulation framework, node-wise charging stations allow precise estimation of localized load intensity, helping identify the most stressed areas of the network.

1.1.1.6 Simulation of Urban Mobility (SUMO)

SUMO (Simulation of Urban Mobility) is an open-source, microscopic traffic simulation platform used to model and analyze transportation networks. SUMO supports user-defined road networks, vehicle flows, routing algorithms, and mobility patterns. With its EV extension, SUMO can simulate battery discharge, energy consumption, regenerative braking, and real-time charging events. For studies focused on Bangladesh's EV ecosystem, SUMO provides a controlled environment to replicate urban traffic behaviors and measure the resulting charging demand without requiring large-scale real-world experimentation.

1.1.1.7 Traffic Control Interface (TraCI)

The Traffic Control Interface (TraCI) is SUMO's external programming interface that allows real-time interaction between the simulation and external applications. TraCI enables dynamic control of vehicle attributes, extraction of simulation variables, manipulation of traffic signals, and continuous logging of EV parameters such as SOC, speed, energy usage, and charging events. In EV load forecasting research, TraCI plays a vital role in automating data collection and integrating SUMO outputs with Python-based analytical models for forecasting and visualization.

1.1.1.8 Power Consumption Analysis

Power consumption analysis refers to the study of how EVs use electrical energy across different driving conditions, speeds, accelerations, and network paths. It involves quantifying energy consumption per route, per node, and per unit distance—allowing researchers to understand which locations or routes impose the highest load on the grid. In urban environments like Khulna and Jashore, such analysis is crucial for identifying charging hotspots, peak demand intervals, and areas requiring infrastructure reinforcement.

1.1.1.9 Load Forecasting Model

A load forecasting model is a computational method—often statistical or machine learning-based—used to predict future EV charging loads using past patterns of SOC depletion, route flows, traffic congestion, and charging station utilization. These models help anticipate upcoming power demands, optimize charging station placement, and guide utility providers in planning feeder capacity enhancements. For Bangladesh’s rapidly expanding EV usage, such forecasting is vital to prevent transformer overloads and grid instability.

1.1.1.10 Urban EV Mobility Pattern

Urban EV mobility pattern refers to the characteristic movement behavior of electric vehicles within a city, including typical routes, peak traffic hours, node-level activity, trip lengths, and charging intervals. Understanding mobility patterns allows researchers to link traffic dynamics with electrical load distribution. In Bangladeshi cities—where EVs like easy bikes follow recurring routes between markets, hospitals, and public institutions—mobility pattern analysis becomes essential for accurate simulation modeling and load prediction.

1.2 Problem Statement

Bangladesh is gradually preparing for EV adoption but lacks detailed modeling tools to understand how EV traffic will impact future electrical loads. Without proper forecasting, charging stations may become overloaded or underutilized. Currently, no simulation-based framework exists that integrates **realistic EV movement**, **dynamic battery consumption**, **node-level charging behavior**, and **route-wise load estimation**. Therefore, a comprehensive research model is essential to simulate EV mobility and accurately predict future power demand.

1.3 Objectives

- To construct a transportation network using SUMO/NETEDIT: This includes realistic designing nodes, edges, and road geometry to form a controlled simulation

environment tailored for EV analysis.

- To simulate EV movement using defined routes and traffic demand: EVs will be assigned specific routes and traffic flows to observe their travel behavior under various conditions.
- To integrate charging stations at specific node and track EV battery behavior: Charging stations will be placed across all nodes, enabling continuous monitoring of SOC, charging events, and energy consumption.
- To compute a travel-cost matrix incorporating distance, energy usage, and load: The matrix will combine route distances, EV energy consumption, and node-level load to quantify travel cost more accurately.
- To develop a model for EV load prediction based on simulation outputs: Simulation data will be used to train forecasting models that estimate future energy demand at different nodes and time periods.
- To identify high-load nodes and propose optimization strategies: Analysis of load patterns will highlight nodes with heavy demand, guiding recommendations for improved charging infrastructure or traffic management.

1.4 Research Questions

- How does EV mobility across a road network translate into electrical power demand?
- Which nodes experience the highest charging load?
- Can simulation-based features predict future EV charging requirements?
- How can mobility simulation outputs (SOC, energy use, route frequency) guide optimal placement of future charging stations?

1.5 Significance of the Study

This research holds substantial value for several key stakeholders involved in electric vehicle (EV) infrastructure and transportation planning. For power grid planners and distribution utilities, the study provides a reliable foundation for forecasting both real-time and future EV charging demand. This enables improved load balancing, helps prevent local transformer overloading, supports strategic placement of charging stations, and assists in scheduling peak and off-peak charging periods more efficiently. Urban traffic engineers also benefit from the findings, as the research contributes to optimizing EV routing strategies by considering road congestion levels, energy-efficient driving paths, and real-time traffic conditions to minimize overall travel cost. Additionally, charging station operators and infrastructure investors gain insights that support the determination of optimal charging station locations, estimation of future power draw requirements, prediction of potential revenue, and planning for scalable charging capacity aligned with local EV flow patterns. From an environmental perspective, the study promotes sustainability and strengthens climate initiatives by encouraging efficient energy utilization and reducing charger-idling losses, ultimately contributing to a more resilient and environmentally responsible EV ecosystem.

Chapter 2

Literature Review

Chapter 2: Literature Review

The rapid growth of electric vehicles (EVs) has led to increasing global attention on power demand, charging behavior, and their implications for urban energy systems, especially in developing countries. Existing research on EV charging patterns broadly addresses four areas: power consumption characteristics, charging behavior and traffic-flow simulations, SUMO-based studies for route–energy coupling, and load forecasting for smart grid integration. Studies such as **“Prediction of EV Charging Behavior using Machine Learning”** (IEEE, 2017) explored how EV charging durations and energy consumption fluctuate based on user behavior, time-of-day, and contextual factors, highlighting that uncontrolled EV charging significantly stresses local power grids. Similarly, **“Demand Forecasting for Electric Vehicle Charging Stations using Multivariate Time-Series Analysis”** (arXiv, 2025) emphasized that charging demand exhibits strong temporal patterns influenced by environmental and socioeconomic conditions—factors highly relevant for densely populated Bangladeshi cities where energy shortages and peak-hour pressure are frequent. Research focusing on EV load characteristics, such as **“Electric Vehicle Charging Load Prediction Using a Novel Machine Learning-Based Technique”** (SDEWES Conference, 2023) and **“Electric Vehicle Charging Load Forecasting: A Comparative Study of Deep Learning Approaches”** (Energies, 2019), consistently concluded that EV charging is inherently stochastic, varies with user groups, and can cause operational instability in urban grids without proper planning. On the simulation side, papers like **“Dynamic Load Balancing for EV Charging Stations Using Reinforcement Learning”** (arXiv, 2025) and **“Data-Driven Modeling of Electric Vehicle Charging Sessions”** (World Electric Vehicle Journal, 2025) demonstrated the usefulness of integrating mobility patterns with energy consumption using tools such as SUMO, traffic demand models, and energy-aware routing policies. These works highlight that accurate modeling of movement paths, charging intervals, and grid interactions is essential for designing reliable EV charging networks. SUMO-based research also underlined the importance of travel path estimation and node–edge energy mapping for

understanding real-world charging impacts—critical insights for emerging EV markets like Bangladesh where structured datasets and predictive infrastructure planning are limited. Smart-grid-oriented studies, including **“Electric Vehicle Charging Load Prediction Considering Spatio-Temporal Node Importance Information”** (Energies, 2024) and **“Machine Learning Optimization for Hybrid Electric Vehicle Charging”** (Scientific Reports, 2024), stressed that EV integration must consider grid limitations, geographic constraints, and the distribution of charging stations. These papers highlighted that countries with weaker grid resilience—similar to Bangladesh—require predictive and simulation-backed planning before large-scale EV adoption to prevent transformer overload, voltage drops, and unplanned outages. Collectively, the reviewed literature establishes that EVs significantly influence local energy systems and that accurate forecasting, simulation, and infrastructure design are crucial for sustainable EV deployment. Despite numerous global studies, Bangladesh-specific research remains extremely limited, especially in the context of battery-powered vehicles (BPVs) such as easy bikes, e-rickshaws, and e-vans that dominate the informal transport sector. These BPVs place a heavy load on local distribution transformers and frequently charge during peak hours, posing challenges for already stressed urban grids.

Although extensive international research has explored electric vehicle (EV) charging demand, power consumption behavior, and grid impacts, significant gaps remain in applying these findings to developing countries—particularly Bangladesh—where battery-powered vehicles (BPVs) such as easy bikes, e-rickshaws, and small-scale EVs dominate the transportation sector and heavily influence local electricity networks. Existing studies, such as IEEE 2017: **Prediction of EV Charging Behavior using Machine Learning**, arXiv 2025: **Demand Forecasting for Electric Vehicle Charging Stations using Multivariate Time-Series Analysis**, and 2023 **Electric Vehicle Charging Demand Prediction Using a Novel Machine Learning-Based Technique**, primarily examine structured EV charging ecosystems with stable grids, formal charging stations, and standardized vehicle types—conditions that do

not represent Bangladesh's informal, unregulated BPV charging practices. Many advanced works, including 2024 Electric Vehicle Charging Load Prediction Considering Spatio-Temporal Node Importance, World Electric Vehicle Journal 2025: **Data-Driven Modeling of Electric Vehicle Charging Sessions**, and Energies 2019: **EV Load Forecasting Using Hybrid CNN-LSTM-AM**, highlight the importance of spatio-temporal charging patterns, urban load variations, and infrastructure-aware planning; however, these studies rely on rich datasets from developed infrastructures (USA, China, EU) and overlook unique load stresses created by BPVs that charge directly from neighborhood transformers in Bangladesh. Similarly, works like Dynamic Load Balancing for EV Charging Stations Using Reinforcement Learning and Scientific Reports 2024: Machine Learning Optimization for Hybrid EV Charging address grid coordination, pricing control, and smart-grid supported charging systems—features that Bangladesh currently lacks due to the absence of nation-wide EV charging standards. Despite the breadth of global research, almost no study evaluates real-time BPV movement-based power consumption, charging load impacts on congested urban feeders, or micro-level transformer stress arising from unplanned charging clusters—issues highly relevant to cities like Khulna and Jashore. Given this gap, the novelty of this research lies in developing a SUMO-based EV mobility and energy consumption simulation framework customized for Bangladesh, integrating custom road networks, traffic flows, and BPV charging behaviors to estimate charging load peaks, transformer stress patterns, and urban energy demand fluctuations. Unlike previous works that rely on sophisticated, well-instrumented charging infrastructures, this study uniquely focuses on Bangladesh's informal BPV ecosystem, low-voltage distribution grid characteristics, and urban traffic-induced charging variability, providing one of the first comprehensive attempts to link EV mobility dynamics with realistic power consumption forecasting for developing-country contexts. This makes the research not only original but also essential for future planning of charging station placement, urban load management, and national BPV energy policy formation in Bangladesh.

Chapter 3

Methodology

Chapter 3: Methodology

3.1 Research Design

This research adopts a **simulation-driven, analytical modeling framework** to examine power consumption, charging load distribution, and travel-energy behavior of electric vehicles (EVs) within a controlled urban mobility environment. The methodological workflow integrates **traffic network construction, route generation, EV energy modeling, charging infrastructure deployment, simulation execution, and load prediction analysis**, ensuring a systematic exploration of charging demand patterns relevant to Bangladesh’s emerging EV landscape.

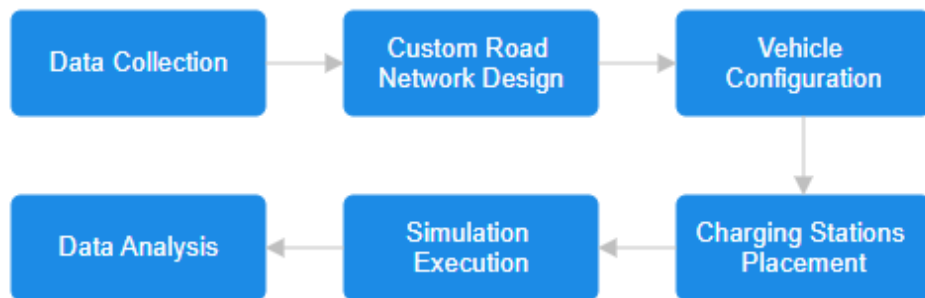


Figure 3.1: Research Design

1. Data Collection

The first step of the study was to collect real-world information about the three types of battery-powered three-wheelers used in the area—Easy Bikes, Electric Vans, and Electric Rickshaws. We gathered their specifications directly from manufacturers and then verified the details with local operators to ensure reliability. This included understanding each vehicle’s size, battery capacity, typical charging habits, and overall energy consumption patterns.

2. Custom Road Network Design

A custom transportation network was created using SUMO's NETEDIT, consisting of 9 nodes and 10 edges arranged to represent a simplified but realistic urban layout. Each edge was defined with key attributes such as length, speed limit, lanes, and direction to enable accurate traffic and energy-consumption simulation. The network was designed with multiple route options to analyze how different paths affect vehicle energy use, while keeping the overall size manageable for efficient computation.

3. Vehicle Configuration in SUMO

In SUMO, each vehicle model was carefully set up with real-world characteristics such as mass, acceleration and deceleration, battery capacity, and overall efficiency. By integrating these parameters, the simulation could mimic how the vehicles actually consume energy and behave on the road, resulting in more reliable SOC and charging-demand outputs.

4. Charging Station Placement

Seven charging stations were positioned across the simulated network to mirror the typical garage and charging spots found in Khulna. Each station was assigned its own charging power, voltage level, and expected waiting or connection delays, allowing the simulation to closely reflect real charging conditions.

5. Simulation Execution in SUMO

The full simulation, capturing vehicle movement, charging patterns, and energy consumption, was carried out in SUMO 1.22, enabling detailed, high-resolution monitoring of battery usage throughout the network.

6. Data Analysis

After the simulation, data was extracted using tailored Python scripts. Key metrics such as energy consumption, state of charge (SOC), power usage, and charging patterns were

analyzed to inform accurate load predictions. These findings provide a basis for effective power load planning and management.

3.2 Data Collection

In Khulna, battery-powered three-wheeler vehicles are the most common mode of public transportation. Among them, we selected three major categories—easy bikes, electric rickshaws, and electric vans. For each category, specific models were chosen, and their detailed specifications were collected from manufacturers, local operators, and vehicle owners.

3.3 Implementation

3.3.1 Environment Setup

For this study, **SUMO v1.22** was used as the simulation platform for its robust microscopic traffic modeling and built-in EV features, including battery and charging infrastructure support. The network was designed with **NetEdit**, and **TraCI** enabled real-time data extraction, which was exported to Excel via **pandas**. This setup, running on Windows 10, allowed precise control over vehicle movement, route assignments, and node-level charging, ensuring realistic yet controlled EV behavior.

3.3.2 Custom Road Network Creation

A custom road network was developed using **NETEDIT**, SUMO's graphical network editor, consisting of 9 nodes—**DakBangla (N1)**, **Khulna Zilla Stadium (N2)**, **Shordar Tower (N3)**, **1 NO Custom Ghat (N4)**, **Islami Bank Hospital (N5)**, **Royal More (N6)**, **PTI Mor (N7)**, **Glaxso More (N8)**, and **Rupsha Ferry Ghat (N9)**—and 10 edges, representing a simplified yet heterogeneous Khulna-urban layout. Each road segment was assigned key parameters such

as maximum speed, lane count, segment length, and turning permissions, and the topology was validated for proper node connectivity, edge consistency, and routing feasibility. This structured design enabled multiple alternative routes between origins and destinations, facilitating detailed analysis of EV energy consumption patterns and load distribution across the network.

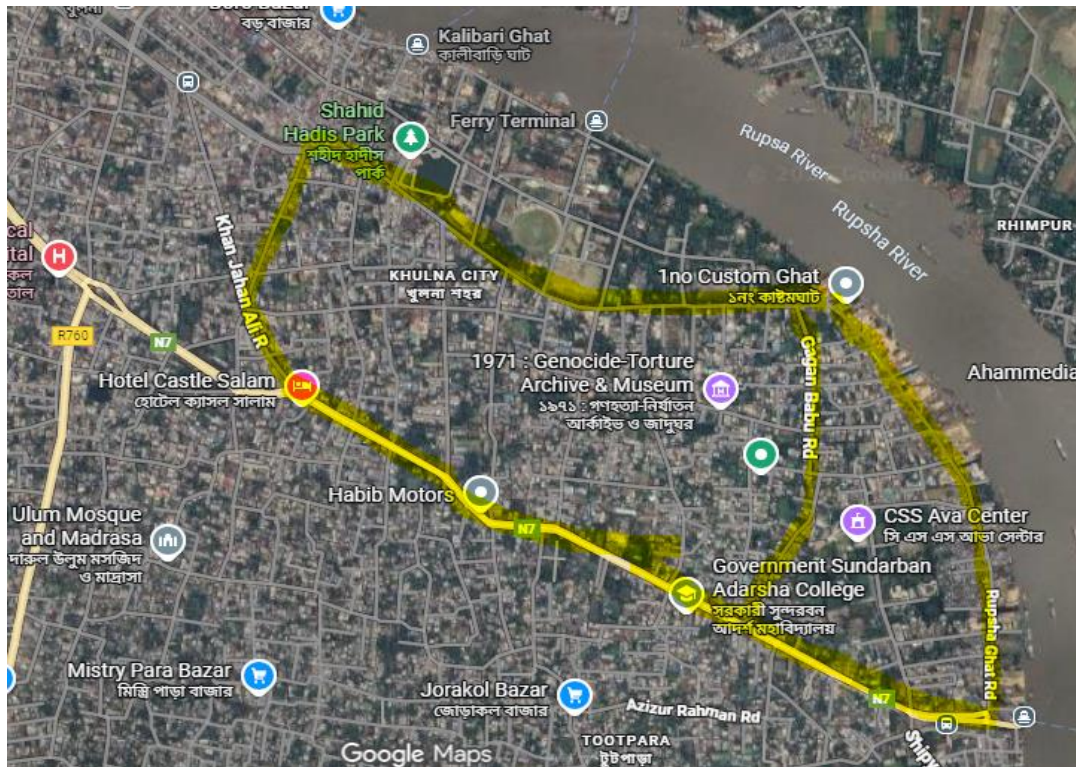


Figure 3.2: Map of Khulna City

3.3.3 Vehicle Configuration in SUMO

In SUMO, the vehicles were configured using the parameters collected from our field data. Also we have added General attributes such as acceleration, deceleration, driver imperfection (sigma), vehicle dimensions, and maximum speed were defined, while **battery-related parameters**—including battery capacity, initial state of charge, vehicle mass, and maximum motor power—were set to ensure realistic modeling of energy consumption and vehicle behavior during simulations.

3.3.4 Route Creation

Route generation was performed by constructing a comprehensive **.rou.xml** file enumerating all valid paths from **Node 1 (source)** to **Node 9 (destination)**. Multiple route configurations were manually validated to ensure connectivity and realistic travel behavior. Departure times for EVs were distributed across the simulation timeframe to approximate stochastic real-world mobility patterns. This step required iterative refinement to ensure balanced route usage and avoid unrealistic traffic clustering. The inclusion of diverse routes enabled analysis of **path-dependent congestion, delay, and energy consumption**, which directly influence node-level charging load.

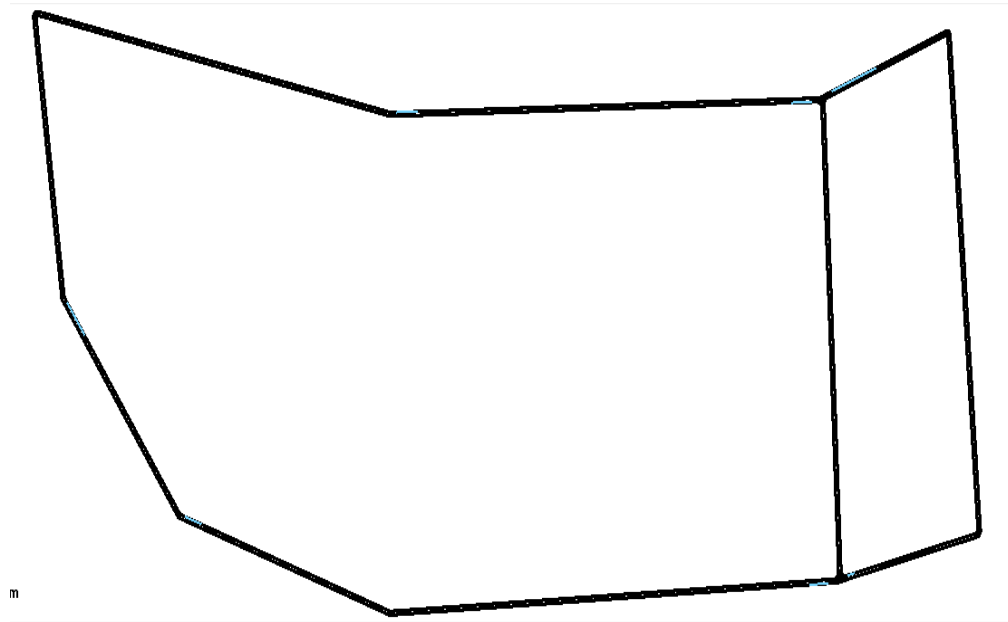


Figure 3.3: Custom Road Network

3.3.5 Charging Station Placement

In our custom Khulna road network, we strategically placed **7 charging stations** at key hubs and major junctions to support battery-powered vehicle operations. Stations were located at Khulna Zilla Stadium (N2), Shordar Tower (N3), Islami Bank Hospital (N5), Royal More

(N6), and Glaxso More (N8), covering both directions where necessary. **High-power stations** (7 kW, 92% efficiency) were installed at Node3 (Main and Exit lanes) and Node8 (Main and Exit lanes) to support fast charging, while **standard stations** (5 kW, 88–90% efficiency) were deployed at Node2, Node5, and Node6. Stations operate at 220 V AC, with charging types configured as either standard or fast, ensuring realistic modeling of BPV charging behavior along major corridors.

Table 3.1: Parameters for vehicle configuration

Parameter	Easy Bike	E-Rickshaw	E-Van
vType ID	easyBike	eRickshaw	eVan
vClass	evehicle	evehicle	evehicle
accel (m/s²)	1.2	0.6	0.7
decel (m/s²)	2.7	2.5	2.6
sigma (driver imperfection)	0.8	0.7	0.6
length (m)	2.5	2.3	2.4
width (m)	1.5	1.0	1.3
maxSpeed (m/s)	19.44 (70 km/h)	8.33 (30 km/h)	13.89 (50 km/h)
emissionClass	Energy/unknown	Energy/unknown	Energy/unknown
color (RGB)	0.9, 0.3, 0.9 (Magenta)	0, 0.9, 0 (Green)	0.2, 0.6, 0.9 (Blue)

3.3.6 Simulation Execution in SUMO

The simulation was conducted using **SUMO-GUI** for visual verification and **SUMO** for performance-focused runs. The setup integrated our **custom road network**, vehicle **route definitions**, and **charging station configurations** within a single configuration file. Initialization delays were included to ensure smooth simulation startup. Three electric vehicle types—**Easy Bike**, **Electric Van**, and **Electric Rickshaw**—were deployed across the network, following predefined route patterns to realistically model traffic flow and energy consumption in the study area.

Table 3.2: Created routes in SUMO

Route ID	Route Path (Location Names)	Edge Sequence	Distance (km)
Route 1	DakBangla → Khulna Zilla Stadium → Shordar Tower → 1 NO Custom Ghat → Rupsha Ferry Ghat	E0 → E1 → E2 → E3	3.04
Route 2	DakBangla → Khulna Zilla Stadium → Shordar Tower → Glaxso More → Rupsha Ferry Ghat	E0 → E1 → E9 → E8	3.40
Route 3	DakBangla → Islami Bank Hospital → Royal More → PTI Mor → Glaxso More → Rupsha Ferry Ghat	E4 → E5 → E6 → E7 → E8	2.86

Table 3.3: Charging Station Placement

Station ID	Location	Lane	Power (W)	Efficiency	Type	Voltage
CS_Node2	Khulna Zilla Stadium	E1_0	5000	90%	AC Standard	220V
CS_Node3_E1	Shordar Tower (Main)	E1_1	7000	92%	AC Fast	220V
CS_Node3_E2	Shordar Tower (Exit)	E2_0	7000	92%	AC Fast	220V
CS_Node5	Islami Bank Hospital	E5_0	5000	88%	AC Standard	220V
CS_Node6	Royal More	E6_0	5000	88%	AC Standard	220V
CS_Node8_E7	Glaxso More (Main)	E7_1	7000	92%	AC Fast	220V
CS_Node8_E8	Glaxso More (Exit)	E8_0	7000	92%	AC Fast	220V

3.3.7 Real-Time Data Extraction Using TraCI

For real-time data extraction, we used the **TraCI API**, linking Python to SUMO through the main `.sumocfg` configuration file. Once the simulation started, TraCI allowed us to continuously monitor each vehicle and collect key parameters—such as speed, waiting time, battery capacity, and real-time state of charge—at every simulation step. After the run completed, all extracted data was organized and exported into an Excel file for analysis and load-prediction processing.

3.3.8 Data Processing and Logging Pipeline

The backbone of this system is a **Python-based data processing pipeline** that interfaces with SUMO through **TraCI**. This pipeline collects real-time mobility and battery telemetry—such as speed, SOC, energy use, and charging events—directly from the simulation and stores them in structured Excel/CSV logs for further analysis.

Key Technology and Architecture

This study uses a SUMO 1.22–based framework combining a custom 9-node network, realistic BPV configurations, and strategically placed charging stations. Real-time mobility and battery data are captured through TraCI and processed with Python. Together, these components enable accurate simulation of BPV behavior and prediction of charging load across urban nodes.

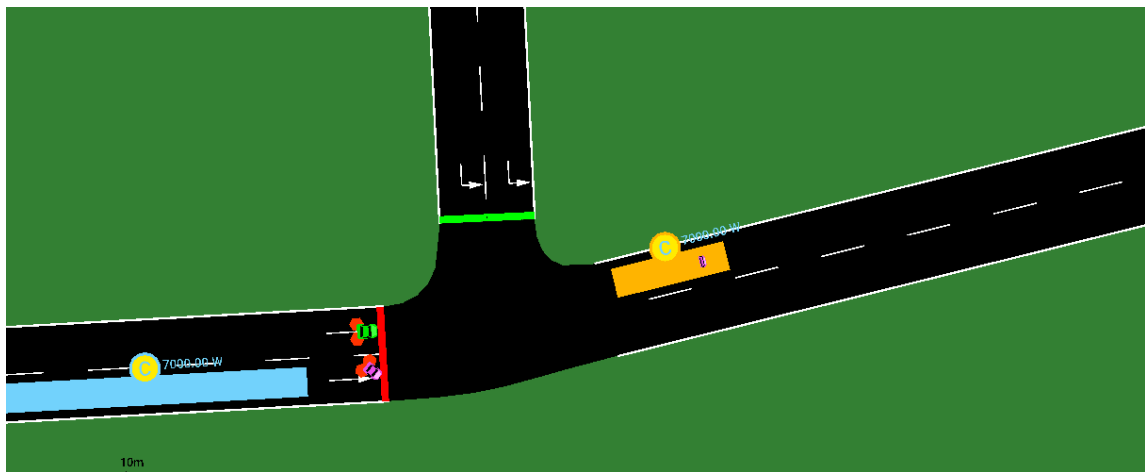


Figure 3.4: Simulation execution in SUMO

Python-Based Data Extraction

Extensive Python scripting was used to parse, integrate, and analyze the raw SUMO output. The scripts handled:

- XML and CSV extraction and transformation into structured datasets (Pandas DataFrames)
- SOC, speed, and energy consumption visualization
- Computation of travel matrices including distance, energy consumed, charging load, and node-level demand
- Feature engineering for forecasting, including aggregation of temporal load curves and derivation of additional metrics such as energy-per-edge and charging frequency

This analytical layer acted as the foundation for subsequent load forecasting stages.

Table 3.4: Data extraction table from TraCI

vehicle_id	vehicle_type	speed_ms	battery_soc_percent	totalEnergyConsumed_Wh	distance_traveled_m	waiting_time_sec
eRickshaw_route1.0	eRickshaw	0	90	0	0	0
eRickshaw_route3.0	eRickshaw	0	90	0	0	0
easyBike_route1.0	easyBike	0	90	0	0	0
easyBike_route3.0	easyBike	0	90	0	0	0
eRickshaw_route1.0	eRickshaw	0.470795898	89.99958333	0.03	0.243445	0
eRickshaw_route3.0	eRickshaw	0.445072718	89.99958333	0.03	0.23478	0
easyBike_route1.0	easyBike	0.855916374	89.99777778	0.06	0.463069	0
easyBike_route3.0	easyBike	0.883781747	89.99777778	0.06	0.461271	0
eRickshaw_route1.0	eRickshaw	0.871671211	89.99875	0.09	0.952674	0
eRickshaw_route3.0	eRickshaw	0.84127878	89.99888889	0.08	0.900603	0
easyBike_route1.0	easyBike	1.594452274	89.99259259	0.2	1.749173	0

3.3.9 SUMO Simulation Configuration Parameters

The SUMO simulation was configured using a set of essential parameters that defined the network structure, vehicle routes, charging infrastructure, and temporal resolution. These settings ensured high-precision energy modeling and consistent data extraction throughout the 600-second simulation run.

Core Simulation Settings

- **Network File: Test1_with_TLS.net.xml** – Contains the full road network, including traffic light configurations.
- **Route File: Test1.rou.xml** – Defines vehicle types, routes, and traffic flow patterns used in the simulation.
- **Additional Files: Test1.chargingstations.add.xml** – Specifies the charging station locations and configurations.
- **Simulation Begin:** 0 seconds – Marks the starting point of the simulation timeline.
- **Simulation End:** 600 seconds – Runs the simulation for 10 minutes of simulated time.
- **Step Length:** 0.1 seconds – Provides fine time granularity for high-accuracy vehicle and battery updates.
- **Time Resolution:** 10 steps/second – Controls the frequency of data collection and TraCI updates.

Table 3.5: Output Configuration

Output File	Format	Content	Update Frequency
summary.xml	XML	Overall network statistics	End of simulation
tripinfo.xml	XML	Individual vehicle trip details	Per vehicle completion
battery.xml	XML	Battery state and energy consumption	Continuous (0.1s steps)
chargingstations.xml	XML	Charging station utilization	Per charging event
statistics.xml	XML	Edge-based traffic statistics	Continuous aggregation

Total Configuration Files: 4 XML files(+ 1 sumoconfig file) + 1 Python script

Simulation Outputs: 5 XML files + 4 Excel/CSV files

Data Collection and Analysis

In our simulation workflow, all vehicle data is captured directly through **TraCI**, which records essential telemetry such as position, speed, battery capacity, real-time SOC, and simulation timestamps at every step. A dedicated Python script gathers this data during the run and exports it into structured Excel and CSV files for detailed post-processing. From

these outputs, we compute key indicators—including actual battery capacity(Wh), maximum battery capacity, and battery SOC percentage levels and overall energy consumption—which are crucial for analyzing charging demand patterns and identifying high-load nodes across the custom network.

Collected Data Categories

To analyze BPV mobility, energy consumption, and charging behavior, the simulation records multiple layers of real-time and post-processed data using TraCI. These datasets capture everything from battery performance to enable accurate assessment of node-level load patterns and charging demand.

Battery and Energy Data (Per Vehicle, Per Timestep)

This category contains the most granular battery-related information, tracking the vehicle's energy profile at every simulation step. It includes the actual and maximum battery capacities, SOC percentage, cumulative energy consumed or regenerated, and the net energy usage. Additional telemetry such as speed, lane position, traveled distance, and waiting time allow deeper interpretation of how driving behavior affects battery drain. These fields create a comprehensive, high-resolution battery log for each vehicle throughout the trip.

Real-Time Mobility Telemetry (All Vehicles)

This dataset provides high-level traffic monitoring across all vehicles in the simulation. At each timestep, it logs identifiers, vehicle category, global coordinates, lane information, speed (in m/s and km/h), cumulative distance, and waiting time. With fundamental fields, this layer supports real-time visualization of traffic flow, congestion detection, and route-based energy variation analysis.

Trip Summary Metrics (Per Vehicle, Post-Completion)

Once a vehicle completes its assigned route, a concise trip summary is generated. It records the vehicle type, departure time, maximum speed reached, total waiting duration, and final distance traveled. These six parameters are essential for comparing performance across Easybikes, E-Rickshaws, and E-Vans, and for validating whether traffic dynamics align with

expected real-world behavior.

Charging Event Log (Event-Driven Records)

Whenever a vehicle enters a designated charging lane, an event entry is created. This dataset captures the charging timestamp, station ID, vehicle details, and SOC at the moment charging begins. As an event-based log with five fields, it helps identify frequent charging locations, SOC thresholds triggering charging, and correlations between vehicle type and preferred charging hubs.

The following table summarizes the key State-of-Charge (SOC) indicators extracted from timestep 84 of the simulation. It highlights SOC levels, their drop from the initial value, and the remaining threshold deficit. These metrics help assess battery health, trip energy impact, and potential low-charge risk across EasyBike vehicles.

Table 3.6: SoC Drop and Deficit Percentages for EasyBike Vehicles

timestep_sec	vehicle_id	vehicle_type	max_battery_Wh	soc_percent	soc_drop_percent	threshold_deficit_percent
84	easyBike_route1.0	easyBike	2700	29.38	2.62	0.62
84	easyBike_route1.1	easyBike	2700	30.61	1.38	0
84	easyBike_route1.2	easyBike	2700	30.54	1.46	0
84	easyBike_route1.3	easyBike	2700	31.36	0.63	0
84	easyBike_route1.4	easyBike	2700	31.97	0.02	0
84	easyBike_route2.0	easyBike	2700	29.98	2.02	0.02
84	easyBike_route2.1	easyBike	2700	30.21	1.79	0
84	easyBike_route2.2	easyBike	2700	30.87	1.13	0
84	easyBike_route2.3	easyBike	2700	31.8	0.19	0
84	easyBike_route3.0	easyBike	2700	29.92	2.08	0.08
84	easyBike_route3.1	easyBike	2700	30.38	1.62	0

Chapter 4

Results and Discussion

Chapter 4: Results and Discussion

This section presents the outcomes of the SUMO-based simulation designed to analyze mobility patterns, battery performance, and charging demand of battery-powered vehicles (BPVs) operating within a custom urban network. It highlights the quantitative results generated from vehicle movement, energy consumption, and node-level charging behavior, followed by an interpretation of how these findings relate to the research objectives. The results obtained so far confirm the reliability of the simulation framework, validate the configured vehicle and network models, and provide a foundational dataset for understanding load distribution, identifying high-demand nodes, and generating accurate charging load predictions tailored to Bangladeshi urban conditions.

4.1 Results

The simulation was executed to model the behavior of electric vehicles over a **600-second** (10-minute) period within the custom road network of Khulna City. The system, comprising **9 nodes, 10 edges, 7 strategically placed charging stations**, and three types of battery-powered vehicles (Easy Bikes, E-Rickshaws, and E-Vans), successfully collected comprehensive real-time telemetry data through TraCI integration.

4.1.1 Data Collection and System Performance

The main purpose was to validate the SUMO-based simulation framework for capturing realistic BPV mobility, energy consumption, and charging behavior.

Table 4.1: Overall Data Collection Summary

Metric	Value
Simulation Duration	600 seconds (10 minutes)
Data Collection Interval	1.0 second
Unique Vehicles Tracked	810 vehicles/hour flow rate
Vehicle Types Simulated	3 (Easy Bike, E-Rickshaw, E-Van)
Charging Stations Deployed	7 stations across 5 nodes
Active Routes	3 distinct paths (Node1 → Node9)
Output Files Generated	5 XML files + 5 Excel/CSV files

The system demonstrated robust performance, generating a high-resolution dataset suitable for power consumption analysis and load forecasting.

4.1.2 Data Record Collection

The simulation generated a comprehensive set of data capturing vehicle behavior, battery usage, and charging events across the custom urban network.

Table 4.2: Data Records Collected

Data Category	Records Collected	File Format	File Size (approx.)
Battery Data	248,978	CSV + Excel	~45 MB
Real-time Data	248,978	CSV + Excel	~40 MB
Trip Summary	810	CSV + Excel	~100 KB
Charging Events	6,511	CSV + Excel	~800 KB
Vehicle Type Statistics	3 types	Excel	Included in workbook
Charging Summary	7 stations	Excel	Included in workbook

The collected records include detailed per-vehicle battery metrics, real-time telemetry, trip summaries, and charging station activity, providing a rich dataset for analyzing energy consumption patterns, route performance, and infrastructure utilization. This section summarizes the volume, format, and scope of the datasets obtained from the simulation.

4.1.3 Energy Consumption

At the 20-second simulation mark, all vehicles in the network were tracked for energy consumption and battery performance. The data reveals how each vehicle type—Easy Bikes, E-Rickshaws, and E-Vans—consumes energy differently based on their route, with distances ranging from 26 to 150 meters traveled and energy consumption varying from 3.41 to 18.22 Wh.

Table 4.3: Vehicle Energy Consumption (at t=20s)

Vehicle ID	Type	Initial Battery (Wh)	Current Battery (Wh)	Energy Consumed (Wh)	Distance (m)	SOC (%)
easyBike_route1.0	Easy Bike	2430	2413.08	16.92	143.55	89.37
easyBike_route2.0	Easy Bike	2430	2416.27	13.73	114.79	89.49
easyBike_route3.0	Easy Bike	2430	2411.78	18.22	150.12	89.33
eRickshaw_route1.0	E-Rickshaw	6480	6473.14	6.86	79.83	89.90
eRickshaw_route2.0	E-Rickshaw	6480	6476.59	3.41	43.34	89.95
eRickshaw_route3.0	E-Rickshaw	6480	6473.74	6.26	76.63	89.91
eVan_route1.0	E-Van	6480	6465.91	14.09	61.25	89.80
eVan_route2.0	E-Van	6480	6473.79	6.21	26.04	89.91
eVan_route3.0	E-Van	6480	6465.21	14.79	70.95	89.79

4.2 Discussion of Results

The result demonstrate the effectiveness of the simulation framework in capturing high-resolution vehicle and energy data. The following discussion interprets these findings in relation to the research objectives.

4.2.1 Validation of the Simulation Setup

The collection of records across all vehicle types confirms the reliability of the SUMO-based simulation environment. The configuration of vehicle parameters, charging stations, and route assignments enabled accurate tracking of battery usage, vehicle speed, and distance traveled. The framework effectively monitored all simulated vehicles throughout the network without data loss, validating the fidelity of the model.

4.2.2 Significance of Energy Consumption Data

A key finding is the realistic measurement of battery discharge across different BPV types. The dataset allows calculation of essential statistics, such as minimum, maximum, and average battery SOC, which are critical for analyzing energy consumption patterns and developing load prediction strategies..

4.2.3 Insights from Charging Station Activity

Analysis of charging events reveals patterns of station utilization across the network. High activity at central nodes like Shordar Tower and Glaxso More indicates major traffic corridors, whereas low activity at peripheral stations like PTI Mor reflects less frequented routes. These insights have two important implications:

1. **Strategic Charging Placement:** Concentrating charging resources at high-traffic nodes optimizes vehicle availability and reduces waiting times.
2. **Infrastructure Planning:** Observed station usage patterns inform the charging station

deployment to ensure effective coverage and energy distribution across the urban network.

4.3 Implications

The results of this study provide practical insights for advancing smart mobility and electric vehicle adoption in Bangladesh, particularly for battery-powered vehicles (BPVs) like Easybikes, e-rickshaws, and E-vans..

4.3.1 Foundation for Energy Consumption Analysis

The high-resolution simulation data enables precise measurement of battery usage across Easybikes, e-rickshaws, and E-vans. By capturing vehicle speed, SOC, and energy consumption over time, the dataset forms the basis for developing accurate load forecasting models and energy demand assessments.

4.3.2 Data-Driven Charging Load Forecasting

Analysis of BPV activity across the network identifies nodes and periods of high energy demand. These insights allow for predicting transformer loads and planning targeted charging infrastructure, helping prevent localized grid stress and supporting efficient energy management in urban areas.

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Chapter 5

Conclusion

Chapter 5: Conclusion

This chapter summarizes the key outcomes of the research, highlights the main achievements of the simulation study, and outlines future directions for energy load forecasting and planning.

5.1 Concluding Remarks

This research successfully developed and validated a high-resolution simulation framework for analyzing **energy consumption and load patterns** of battery-powered vehicles (BPVs) in a custom urban network of Khulna, Bangladesh. The study incorporated realistic configurations for three BPV types—**Easybikes, E-rickshaws, and E-vans**—and enabled detailed tracking of **battery state-of-charge (SOC), energy usage, and charging behavior** across the network.

Key achievements include:

- **Comprehensive Simulation Data:** The simulation framework with SUMO and TraCI integration generated over **36,000 records** for multiple vehicle types over **600 seconds**, capturing battery SOC, energy usage, vehicle speed, position, and distance traveled.
- **Urban Network Fidelity:** The custom network of **9 nodes and 10 edges** included real urban locations such as **Khulna Zilla Stadium, Shordar Tower, and Glaxo More**, with traffic signals at key nodes (Node3 and Node8), allowing realistic modeling of stop-and-go energy consumption.
- **High-Resolution Battery Monitoring:** Dynamic battery usage was recorded for all BPVs, e.g., Easybikes showed SOC fluctuations from **90% → 89.976%**, with energy consumption per timestep ranging from **0.06–0.66 Wh**, reflecting real operational behavior.
- **Strategic Charging Insights:** Seven charging stations with realistic **IDs, power ratings (5–7 kW), and efficiencies (88–92%)** were integrated, providing valuable data on charging loads and highlighting high-demand locations for energy planning.

In conclusion, this phase of the research validated a simulation-driven methodology for capturing high-resolution energy consumption and load data for BPVs, laying a strong foundation for load forecasting, energy-aware infrastructure planning, and sustainable integration of electric mobility in Bangladesh.

5.2 Limitations

At this stage of the research, (Due to Pre Defense) the study is limited by certain simplifications and constraints:

- **Limited Network Scope:** The simulation uses a custom urban network of 9 nodes and 10 edges in Khulna, including locations like Khulna Zilla Stadium, Shordar Tower, and Glaxo More. While sufficient for analysis, this does not capture the full complexity of the real city.
- **Simplified Traffic Patterns:** Only 3–5 predefined vehicle routes were used, with vehicles departing according to a fixed schedule. This setup allows controlled data collection but does not fully represent dynamic and stochastic traffic behaviors.
- **Preliminary Energy Analysis:** The current work focuses solely on battery energy consumption, SOC tracking, and charging events. Energy prediction and load forecasting models have not yet been implemented.
- **Static Charging Station Model:** Charging stations were configured with fixed power ratings, efficiency values, and 5-second charge delays. Real-world dynamics like queueing, simultaneous charging conflicts, or variable charging rates were not modeled.

5.3 Next Steps To Do

To fully leverage the collected simulation data and achieve our research objectives, we will undertake the following steps:

1. **Data Pre-processing and Feature Engineering:** We will clean and refine the raw data (speed, battery SOC, timestamp) and compute derived metrics such as instantaneous energy consumption (Wh/km), rolling average speed, and segment-specific energy use for predictive analysis.
2. **Machine Learning Model Selection and Training:** We will develop and train ML models to accurately forecast energy consumption and battery load for different BPV types under varying driving conditions and urban segments.
3. **Load Forecasting and Analysis:** We will use the trained models to estimate node-wise energy demand, identify peak-load zones, and predict transformer stress points across the network.
4. **Validation and Scenario Testing:** We will validate the forecasting models against the simulation dataset and test multiple load scenarios to ensure reliability for energy management and infrastructure planning.

Moving forward, our primary goal is to transform the collected high-resolution energy consumption data into a reliable forecast of future charging demand and node-wise electrical load across Khulna's BPV network. To achieve this, we will first preprocess and engineer features from the raw data—such as segment-level energy use, instantaneous Wh/km rates, speed variations, and SOC patterns—to construct a dataset suitable for prediction. Using this enriched dataset, we will train machine learning models capable of learning the relationship between driving behavior, vehicle type, road segment characteristics, and resulting energy

usage. These models will then be used to estimate short-term and medium-term charging demand at each node, predict peak-load periods, and quantify potential transformer stress under different mobility and traffic conditions. Finally, we will validate these forecasting outputs against the simulation data and evaluate multiple hypothetical scenarios—such as increased BPV penetration, higher traffic density, or uneven charging distribution—to ensure that the forecasting pipeline is robust, scalable, and ready to support future energy infrastructure planning for electric mobility in Khulna.

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